MyDakota HIS

Health Intelligence System

Childhood Disease Dashboard

COMPLETE USER GUIDE & DOCUMENTATION

www.mydakotahis.com

Covering:

Dashboard Home • Data Explorer

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1. Introduction to MyDakota HIS

MyDakota HIS (Health Intelligence System) is a publicly accessible, browser-based surveillance dashboard operated by the North Dakota Department of Health and Human Services (ND HHS). It monitors and visualizes the spread of childhood vaccine-preventable diseases across all 53 counties in North Dakota. The platform aggregates confirmed case data, vaccination coverage rates, and population-level risk assessments to provide public health officials, researchers, policymakers, educators, and members of the general public with a clear and up-to-date picture of childhood disease activity in the state.

1.1 Purpose and Scope

The primary purpose of MyDakota HIS is to provide real-time and historical surveillance data for childhood diseases that are largely preventable through vaccination. By combining case counts, immunization records, and geographic data, the platform enables users to:

- Identify counties with elevated disease risk or low vaccination coverage.
- Track disease trends over multiple years (2022 through 2026).
- Compare vaccination rates across all 53 North Dakota counties.
- Download raw datasets for independent research and analysis.
- Understand the relationship between immunization gaps and disease outbreaks.

1.2 Diseases Tracked

The platform tracks the following five childhood diseases:

Disease	Description
Measles	A highly contagious viral infection preventable by the MMR (Measles-Mumps-Rubella) vaccine. The dashboard tracks confirmed cases from 2025 to 2026, reflecting a real outbreak in North Dakota.
Hepatitis A	A liver disease caused by the Hepatitis A virus, spread through contaminated food or water. Preventable by the Hepatitis A vaccine.
Rotavirus	A leading cause of severe diarrhea in young children. Preventable by the Rotavirus vaccine.
RSV	Respiratory Syncytial Virus — a common respiratory virus that can be severe in infants and young children.
Influenza	The seasonal flu — preventable in large part through annual flu vaccination.

1.3 Data Sources

The data displayed on MyDakota HIS comes from the following sources:

- North Dakota Department of Health and Human Services (ND HHS) — confirmed case reports for Measles (2025–2026 outbreak data is real).
- ND Immunization Information System (NDIIS) — vaccination coverage rates by county and year.

All Measles case data reflects actual 2025–2026 confirmed case counts. Hepatitis A, Rotavirus, RSV, and Influenza datasets are derived from ND HHS surveillance records. Vaccination coverage data spans 2022–2025.

1.4 Who Should Use This Guide

This user guide is designed for anyone who visits www.mydakotahis.com, including:

- Public health officials and epidemiologists monitoring disease trends.
- Researchers and academics studying childhood vaccination coverage.
- Healthcare providers seeking county-level data.
- Journalists and policy analysts reporting on public health.
- Members of the general public interested in local disease surveillance.
- Students and educators using real-world health data for learning.

No technical background is required to use the dashboard. This guide explains every feature from start to finish.

2. Website Overview & Navigation

MyDakota HIS is structured around two primary pages accessible from any device with a modern web browser. No login or account is required to access any feature.

2.1 Site Structure

Page	Description
Dashboard Home (index.html)	The main page at www.mydakotahis.com — provides an interactive disease surveillance dashboard with maps, charts, stat cards, and detailed disease-by-disease analysis tabs.
Data Explorer (data.html)	Located at www.mydakotahis.com/data — a dedicated page for browsing, filtering, searching, and downloading the raw surveillance datasets that power the dashboard.
Contact Page	Located at www.mydakotahis.com/contact — for reaching the MyDakota HIS team with questions or feedback.

2.2 The Header (Persistent Navigation Bar)

The dark navy header appears at the top of every page and remains fixed (sticky) as you scroll, so navigation is always accessible. It contains:

Logo / Home Link	Clicking the red shield icon or the text "MyDakota HIS" on the far left always returns you to the Dashboard Home page. The subtitle below reads "Health Intelligence System – Childhood Disease Dashboard."
Home Link	Navigates to the Dashboard Home page (www.mydakotahis.com).
Data Link	Navigates to the Data Explorer page (www.mydakotahis.com/data). When you are on this page, the link is highlighted in a red-tinted background to indicate you are in the active section.
Contact Link	Navigates to the Contact page for submitting queries or feedback.

2.3 Footer

A footer appears at the bottom of both pages in the same dark navy color as the header. It includes attribution and data source acknowledgment, identifying the North Dakota Department of Health and Human Services as the data authority.

3. Dashboard Home Page (www.mydakotahis.com)

The Dashboard Home page is the central hub of MyDakota HIS. It provides an at-a-glance summary of all tracked diseases, and allows users to drill into detailed disease-specific data through an organized tab system. The page is organized from top to bottom into the following zones: Filter Bar, Summary Stat Cards, Disease Tabs, and the main Dashboard Content area (charts, maps, tables).

3.1 The Disease Tab Bar

Directly beneath the header is a row of clickable disease tabs. These tabs allow you to switch the dashboard view between different diseases and the Methodology section. The available tabs are:

Tab	Description
Overview	The default view when you first load the page. Shows a summary across all diseases.
Measles	Focuses the dashboard on Measles case data, vaccination rates, and risk scores. Indicated by a red underline when active.
Hepatitis A	Focuses the dashboard on Hepatitis A data. Indicated by an orange underline when active.
Rotavirus	Focuses the dashboard on Rotavirus data. Indicated by a green underline when active.
RSV	Focuses the dashboard on Respiratory Syncytial Virus (RSV) data. Indicated by a blue underline when active.
Methodology	Opens an explanation of how the dashboard calculates risk scores, processes data, and the definitions used throughout the platform.

Note: *The active tab is highlighted with a colored underline. Clicking a tab instantly switches all charts, maps, and stat cards to the selected disease.*

3.2 The Filter Bar

Beneath the tab bar sits a white rounded filter bar. This bar allows you to refine the data being displayed across the entire dashboard. It contains:

3.2.1 Year Filter (Dropdown)

A dropdown menu labeled "Year" lets you select a specific year (2022, 2023, 2024, 2025, or 2026) or "All Years" to include all available data. Changing this filter instantly updates all charts and statistics on the dashboard.

3.2.2 County Filter (Dropdown)

A dropdown labeled "County" lets you narrow the dashboard view to a specific North Dakota county. By default it shows "All Counties." Selecting a county updates case totals, charts, and maps to reflect only that county's data.

3.2.3 Active Filter Badge

When a filter is applied, a small navy badge appears on the right side of the filter bar indicating the current filter combination (e.g., "2025 • All Counties"). This provides a quick confirmation of what filters are currently active.

3.2.4 How to Reset Filters

To clear all filters, return each dropdown to its default "All" value. There is no single "Reset All" button on the filter bar itself — each filter must be individually returned to its default.

3.3 Summary Stat Cards

Below the filter bar, four colored stat cards display at-a-glance metrics for the currently selected disease tab. Each card features:

- A colored top border matching the disease color (e.g., red for Measles, orange for Hepatitis A, green for Rotavirus, blue for RSV).
- A label describing the metric (e.g., "Total Cases," "Counties Affected," "Critical Risk Counties," "High Risk Counties").
- A large number in bold serif font showing the current value.
- A small sub-label providing context, such as the year range or description of the metric.

These cards update dynamically based on the active tab and any applied filters. For example, when the Measles tab is active, the cards show: Total Measles Cases, Counties Affected, Critical Risk Counties, and High Risk Counties.

3.4 The Overview Tab

When you first load the dashboard (or click the "Overview" tab), you see a multi-disease summary view. This view contains:

3.4.1 Overview Stat Cards

The four stat cards at the top show aggregate data across all tracked diseases, such as total confirmed cases across all diseases, number of counties with at least one reported case, and an overall vaccination coverage indicator.

3.4.2 Multi-Disease Charts

The overview section typically contains combined charts comparing disease trends, allowing users to see at a glance which diseases are most active across different years. These may include line charts comparing annual case totals across diseases, or bar charts showing county-level disease burden.

3.5 The Measles Tab — Detailed Walkthrough

The Measles tab is the most detailed section of the dashboard, reflecting actual confirmed outbreak data from 2025–2026. It contains multiple sub-sections and visualizations. Click the "Measles" tab in the tab bar to activate it.

3.5.1 Measles Stat Cards

The four stat cards at the top refresh to show Measles-specific metrics:

- Total Cases — the total number of confirmed Measles cases across the selected filters (e.g., 60 total across 2025–2026).
- Counties Affected — how many of the 53 counties have reported at least one case.
- Critical Risk Counties — counties with the most severe combination of low vaccination rates and high case counts (shown in red).
- High Risk Counties — counties with elevated but not critical risk levels (shown in orange).

3.5.2 Monthly Case Trend Chart

A dual-line area chart shows Measles case counts by month, with separate lines for 2025 and 2026. This allows users to:

- See which months saw the largest surges (e.g., the peak activity in May 2025 with 34 cases, and February 2026 with 24 cases).
- Compare seasonal patterns between the two outbreak years.
- Identify whether the outbreak appears to be growing, declining, or stabilizing.

The 2026 line is rendered as a dashed line to visually distinguish it from the solid 2025 line. Each year is color-coded (red for 2025, orange for 2026).

3.5.3 County Case Comparison Chart

A horizontal bar chart shows the top counties by number of confirmed Measles cases, split by year (2025 in red, 2026 in orange). This makes it easy to see:

- Which counties experienced the most cases (e.g., Pembina with 21 cases in 2026, Williams with 18 in 2025, Grand Forks with 10 in 2025).
- How the geographic epicenter of the outbreak shifted between years.
- Whether new counties emerged as hotspots in 2026 compared to 2025.

3.5.4 State MMR Vaccination Rate Trend

A line chart tracks the statewide MMR (Measles-Mumps-Rubella) vaccination coverage rate from 2022 to 2025. This is critical context for understanding the outbreak: the data shows a steady decline from 92.23% in 2022 to 89.71% in 2025, falling below the 95% herd immunity threshold required to prevent community spread of Measles.

> **Note:** *The 95% herd immunity threshold for Measles is globally recognized. North Dakota's statewide rate falling below this threshold is directly correlated with the 2025–2026 outbreak.*

3.5.5 Lowest MMR Coverage Counties Chart

A horizontal bar chart shows the 10 counties with the lowest MMR vaccination rates. Counties below 70% are highlighted in red (indicating critical risk); those between 70–85% are in orange. For example, Sheridan County at 57.14% and Grant County at 59.09% represent the most severely under-vaccinated areas in the state. This chart helps users immediately identify where vaccination outreach efforts are most urgently needed.

3.5.6 Disease Risk Score Chart

A horizontal bar chart displays a calculated composite risk score (on a scale of 0–10) for the top 10 highest-risk counties. The risk score is derived from a combination of factors including MMR vaccination rate, number of confirmed cases, and population demographics. Color coding indicates risk level:

- Red bars = Critical risk (score above 4.0) — e.g., Sheridan (4.2) and Grant (4.0).
- Orange bars = High risk (score 2.5 to 4.0) — e.g., Billings (3.23), Logan (2.89).
- Yellow bars = Moderate risk (score 1.5 to 2.5) — e.g., Wells (2.06), Dunn (1.75).

Hovering over a bar in the tooltip shows the risk level label (Critical, High, Moderate, etc.) in addition to the numeric score.

3.5.7 Risk Distribution Doughnut Chart

A doughnut (ring) chart summarizes how North Dakota's 53 counties are distributed across risk levels. The segments are color-coded to match the risk categories used throughout the dashboard:

- Red = Critical
- Orange = High
- Yellow = Moderate
- Green = Minimal Risk
- Gray = Insufficient Data

The legend appears to the right of the chart. This visualization provides an immediate sense of how widespread the risk is across the state.

3.5.8 Population-Adjusted Risk Chart

A horizontal bar chart shows population-adjusted risk scores for the top 10 counties. Unlike the disease risk score which is based primarily on vaccination rates and case counts, this chart weights risk by the size of the susceptible population. Counties with large populations and low coverage may appear here even if they had fewer confirmed cases, because the potential for rapid spread is greater. Colors reflect risk level using a purple palette (darker = more severe).

3.5.9 Susceptible Population Chart

A horizontal bar chart shows the estimated number of susceptible individuals (those not vaccinated or with insufficient immunity) in the top 10 counties by susceptible population. Two bars are shown side by side for each county:

- Purple bar = Estimated total susceptible population (all ages).
- Light purple bar = Estimated susceptible population under 18 years old.

This chart is particularly valuable for understanding where unvaccinated children are most concentrated, as they are at highest risk for severe Measles infection.

3.5.10 Measles County Map

An interactive map of North Dakota appears in the Measles tab, displaying all 53 counties color-coded by the number of confirmed Measles cases. The map uses a Leaflet-based tile map with CartoDB light-style tiles.

Reading the Map Colors

Map Color	Meaning
Dark Red (16+ cases)	County with very high confirmed case count.
Light Red (6–15 cases)	County with moderate-to-high confirmed case count.
Amber (2–5 cases)	County with a low-to-moderate case count.
Light Orange (1 case)	County with a single confirmed case.
Light Gray (0 cases)	County with no confirmed cases in the selected period.

Map Controls

Above or beside the map, three toggle buttons allow you to switch the data shown:

- 2025 — Shows only 2025 confirmed case counts per county.
- 2026 — Shows only 2026 confirmed case counts per county.

- All — Combines 2025 and 2026 cases for a cumulative view.

County Tooltips

Hovering your mouse over any county on the map displays a tooltip showing:

- County name (e.g., "Williams County").
- Cases shown for the currently selected year toggle.
- A breakdown showing both "2025: X" and "2026: Y" case counts side by side.

Map Navigation

The map supports standard zoom controls (+ and - buttons in the top-left corner). Click and drag to pan across the state. Scroll-wheel zoom is disabled by default to avoid accidental zooming when scrolling the page.

3.6 The Hepatitis A Tab

Clicking the "Hepatitis A" tab (with orange accent) switches the dashboard to display Hepatitis A surveillance data. The layout mirrors the structure of the Measles tab and includes:

- Stat cards showing total Hepatitis A cases, counties affected, and risk breakdown.
- Monthly trend charts comparing case counts by year.
- County-level bar charts showing the most affected counties.
- Vaccination coverage charts for Hepatitis A vaccine.
- Risk score charts using the same scoring methodology as Measles.
- An interactive county map color-coded by Hepatitis A case counts.

All visualizations on this tab are filtered to Hepatitis A data only. The data spans 2022 to 2026 and is sourced from ND HHS records.

3.7 The Rotavirus Tab

Clicking the "Rotavirus" tab (green accent) displays Rotavirus surveillance data. Rotavirus primarily affects children under 5 years old and is the leading cause of severe diarrhea in this age group. The tab includes the same types of visualizations as the Measles and Hepatitis A tabs:

- Stat cards with Rotavirus-specific case and risk metrics.
- Monthly trend lines and county bar charts.
- Rotavirus vaccination coverage trend by year and county.
- Disease risk and population risk scores by county.
- An interactive map showing Rotavirus case distribution across North Dakota.

3.8 The RSV Tab

The RSV tab (blue accent) presents surveillance data for Respiratory Syncytial Virus. RSV is a common cause of respiratory illness in infants and young children, and can lead to hospitalization in severe cases. The tab provides:

- Stat cards showing RSV-specific totals and risk ratings.
- Trend charts and county comparisons for RSV cases across 2022–2026.
- RSV vaccination coverage data by county.
- Risk scoring and population susceptibility charts.
- An interactive map displaying RSV activity by county.

3.9 The Methodology Tab

The Methodology tab provides a transparent explanation of how the dashboard works, including:

3.9.1 Risk Score Calculation

The dashboard computes a composite disease risk score for each county on a scale of 0 to 10 using a weighted formula that considers:

1. MMR (or disease-specific) vaccination coverage rate — lower coverage increases the score.
2. Number of confirmed cases in recent years — more cases increase the score.
3. Population density and size of the under-18 demographic — larger susceptible child populations increase the score.
4. Proximity to counties with active outbreaks — geographic adjacency adds a risk modifier.

3.9.2 Risk Level Thresholds

Risk Level	Definition
Critical	Risk score above 4.0. Immediate public health attention recommended.
High	Risk score between 2.5 and 4.0. Elevated risk requiring monitoring.
Moderate	Risk score between 1.5 and 2.5. Some concern; routine surveillance recommended.
Minimal Risk	Risk score below 1.5. Well-vaccinated counties with few or no cases.
Insufficient Data	Not enough data to calculate a meaningful risk score.

3.9.3 Data Update Frequency

The dashboard reflects the most recent data available from ND HHS and NDIIS. Vaccination coverage data is typically updated on an annual basis. Confirmed case data (particularly Measles) may be updated more frequently during active outbreaks.

4. Data Explorer Page (www.mydakotahis.com/data)

The Data Explorer is a dedicated page that allows users to browse, search, filter, sort, and download the raw surveillance datasets that power the Dashboard. This page is ideal for researchers, analysts, and anyone who wants to work with the underlying data directly.

4.1 Page Hero Banner

At the top of the Data Explorer page, a dark navy hero banner displays key statistics about the available data:

Stat	Meaning
Total Records	The total number of rows across all datasets combined. This figure updates dynamically as datasets are loaded (shown as "—" briefly while the page initializes).
5	5 separate datasets are available for exploration.
53	53 North Dakota counties are represented in the data.
2022–2026	The full date range covered by the surveillance data.

Beneath the stats, a paragraph describes the data sources: Measles data reflects real 2025–2026 confirmed case counts from ND HHS, while all other data originates from ND HHS and the ND Immunization Information System (NDIIS).

4.2 Dataset Selector Cards

Below the hero banner, a row of six clickable dataset cards allows you to choose which dataset to explore. Each card displays:

- A color-coded icon representing the disease.
- The dataset name.
- A meta line showing the number of records and fields (e.g., "1,084 records · 5 fields").
- A badge indicating whether it is Case Data or Coverage Data.

The six available datasets are:

Dataset	Details
Measles Cases	9 records · 5 fields · Real confirmed case data from 2025–2026. Color: Red.
Hepatitis A Cases	1,084 records · 5 fields. Simulated/modeled data based on ND HHS records. Color: Orange.
Rotavirus Cases	1,084 records · 5 fields. Color: Green.
RSV Cases	1,084 records · 5 fields. Color: Blue.
Influenza Cases	1,084 records · 5 fields. Color: Purple.
Vaccination Coverage	1,084 records · 6 fields. County-level vaccination coverage rates by disease and year. Color: Dark Purple.

4.2.1 Selecting a Dataset

Click any dataset card to load that dataset into the table below. The selected card is highlighted with a colored border and a subtle glow effect matching the dataset's color. When a new dataset is selected, the table header columns, rows, and available filter options all update automatically to reflect that dataset's structure.

> **Note:** *Selecting a new dataset resets all active filters and returns to page 1 of the table.*

4.3 Data Table Fields by Dataset

Each dataset has its own set of columns (fields). Below are the fields available in each dataset:

4.3.1 Case Datasets (Measles, Hepatitis A, Rotavirus, RSV, Influenza)

Field	Description
Year	The calendar year the case was reported (2022–2026).
County	The North Dakota county where the case was reported (one of 53 counties).
Cases	The number of confirmed cases for that county and year combination.
Gender	The gender of the patient(s): Male or Female. Displayed as a colored badge in the table.
Race	The race/ethnicity of the patient(s): White, Black American, Hispanic, or Native American.

4.3.2 Vaccination Coverage Dataset

Field	Description
Year	The calendar year (2022–2025).
County	The North Dakota county.
Disease	The disease for which coverage is measured (Measles, Hepatitis A, Influenza, Rotavirus, RSV).
Coverage_Pct	The vaccination coverage rate as a percentage (0–100%). Displayed as a color-coded pill in the table.
Vaccinated	Estimated number of vaccinated individuals in that county.
Population	Total population of the county for that year.

4.4 Controls Bar

Above the data table, a white rounded controls bar provides all the tools you need to filter and search the selected dataset. Each control operates in real time — the table updates as soon as you change any value.

4.4.1 Search Box

A text input labeled "Search" allows you to type any keyword and instantly filter the table to rows matching that text. The search checks across multiple columns simultaneously (county name, race, gender, disease name). For example, typing "Williams" will show only rows where Williams County appears. Typing "Native" will show only rows with Native American race.

> **Note:** *The search is case-insensitive. You do not need to capitalize county names or other values.*

4.4.2 Year Filter (Dropdown)

A dropdown labeled "Year" allows you to filter records to a specific year: 2022, 2023, 2024, 2025, or 2026. The default is "All Years." Select a specific year to narrow results to cases or coverage data from that calendar year only.

4.4.3 County Filter (Dropdown)

A dropdown labeled "County" lists all counties present in the currently selected dataset. Selecting a county filters the table to show only records for that county. The county list populates dynamically based on the active dataset, so it always reflects the counties actually in the data.

4.4.4 Gender Filter (Dropdown)

Available for all Case datasets (Measles, Hepatitis A, Rotavirus, RSV, Influenza). Options are "All," "Male," and "Female." Selecting a gender filters to records matching that gender. Gender data is displayed in the table as a color-coded badge: blue for Male, pink for Female.

4.4.5 Race Filter (Dropdown)

Available for all Case datasets. Options are "All," "White," "Black American," "Hispanic," and "Native American." Selecting a race category filters the table to records matching that race. Race is also displayed as a badge in the table.

4.4.6 Disease Filter (Dropdown)

This filter is shown only when the Vaccination Coverage dataset is active. It allows you to filter coverage records to a specific disease (Measles, Hepatitis A, Influenza, Rotavirus, or RSV). This is useful for comparing vaccination rates across counties for a single disease.

4.4.7 Clear Button

A "Clear" button on the right side of the controls bar resets all filters (search text, year, county, gender, race, disease) back to their defaults and returns you to page 1 of the table. Use this button whenever you want to start fresh and see all records.

4.4.8 Download Button

A "Download" button opens the Download Panel (described in Section 4.6) where you can export data in multiple formats.

4.5 Results Info Bar

Between the controls bar and the data table, a thin results bar displays:

- "Showing X of Y records" — shows how many records match the current filters out of the total in the dataset.
- Active Filter Chips — small navy rounded badges appear for each active filter, showing what is currently applied (e.g., "Year: 2025", "County: Cass", "Gender: Female"). Each chip has a small "x" icon that you can click to remove that specific filter without clearing all others.

4.6 The Data Table

The main data table displays the filtered records in rows and columns. Key features of the table include:

4.6.1 Column Sorting

Every column header in the table is clickable and sortable. Click a column header once to sort the table in ascending order by that column. Click it again to sort in descending order. A small chevron icon next to the column name indicates the current sort direction (up arrow = ascending, down arrow = descending). Sorted columns display the chevron in blue to distinguish them from unsorted columns.

4.6.2 Row Striping and Hover Highlighting

Table rows alternate between white and a very light cream color for easier reading. When you hover your mouse over any row, that row subtly highlights to help you track it across columns.

4.6.3 Visual Badges

Certain data fields are rendered with special visual treatments for clarity:

- Gender — displayed as a rounded pill badge: blue background for "Male," pink background for "Female."
- Race — displayed as a rounded pill badge with a subtle light background.
- Cases — for Case datasets, a small mini bar chart within the cell visually represents the case count relative to others in the current view, making it easy to compare magnitudes at a glance.
- Coverage_Pct — for the Vaccination Coverage dataset, coverage percentages are shown as colored pills: green for high coverage (above herd immunity threshold), yellow for moderate, red for low.

4.6.4 Horizontal Scrolling

On smaller screens or when many columns are present, the table supports horizontal scrolling. A scroll bar appears at the bottom of the table area. On touch devices, you can swipe left or right within the table to see all columns.

4.7 Pagination

The data table is paginated to keep load times fast and the interface manageable. Pagination controls appear at the bottom of the table in three parts:

4.7.1 Page Information

On the left side, a text label shows "Page X of Y" — the current page number and total number of pages for the current filtered dataset.

4.7.2 Page Navigation Buttons

In the center, a row of numbered page buttons allows direct navigation to any page. A left arrow ("<") button goes to the previous page, and a right arrow (">") button goes to the next page. The

current page button is highlighted in navy. If there are many pages, ellipsis ("...") buttons appear between the page number buttons to save space, and clicking them is disabled.

4.7.3 Rows Per Page Selector

On the right side, a small dropdown labeled "Rows per page" lets you control how many records are shown per page. Options are 25, 50 (default), 100, and 250. Selecting a larger number shows more records per page but may require more scrolling. The page resets to page 1 whenever you change this value.

> **Note:** *When you change the page, the page automatically scrolls up to the controls bar so you don't have to scroll manually.*

4.8 Download Panel

Clicking the "Download" button in the controls bar reveals the Download Panel, which offers three download options:

CSV (Full Dataset)	Downloads the complete selected dataset as a CSV (Comma-Separated Values) file. This file is compatible with Microsoft Excel, Google Sheets, LibreOffice, and any data analysis tool. The file is named after the dataset (e.g., Measles_Cases.csv).
JSON (Full Dataset)	Downloads the complete dataset as a JSON (JavaScript Object Notation) file. JSON is a structured data format commonly used by developers and data scientists. The file is named after the dataset (e.g., Measles_Cases.json).
Filtered CSV	Downloads only the rows currently visible after applying filters. The filename includes "_filtered" to indicate it is a subset (e.g., Measles_Cases_filtered.csv). This is useful when you want to export only the data relevant to your current research question.

Clicking any download option triggers an immediate file download through your browser. No email or account is required. The downloaded file saves to your default Downloads folder.

4.9 Measles Insight Charts (Special Section)

When the Measles Cases dataset is selected in the Data Explorer, an additional special section of embedded charts appears below or alongside the data table. These charts are specific to Measles and provide the same analytical context as found on the Dashboard Home page. They include:

- County Cases Chart — a horizontal bar chart showing Measles cases by county for 2025 vs. 2026.

- Annual Totals Chart — a vertical bar chart showing total confirmed Measles cases per year from 2022 to 2026. Years 2022–2024 show zero cases (no outbreak), 2025 shows 36 cases, and 2026 shows 24 cases (through February 2026).
- MMR Trend Line — a line chart tracking the state-level MMR vaccination rate from 2022 to 2025, showing a consistent downward trend from 92.23% to 89.71%.

These embedded charts make it convenient to reference analytical context while exploring the raw data table.

5. Step-by-Step Workflows

This section provides practical step-by-step instructions for common tasks you might want to perform on MyDakota HIS.

5.1 How to Check Measles Activity in a Specific County

5. Go to www.mydakotahis.com (the Dashboard Home page).
6. Click the "Measles" tab in the tab bar beneath the header.
7. In the Filter Bar, open the "County" dropdown and select the county you are interested in (e.g., "Williams").
8. The stat cards update immediately to show Measles case totals for Williams County.
9. Scroll down to view the County Case Comparison chart — your selected county will be highlighted.
10. View the interactive map to see Williams County highlighted based on its case count.
11. Hover over the map county to see the tooltip showing 2025 vs. 2026 case counts.
12. To reset, return the County dropdown to "All Counties."

5.2 How to Find the Most Under-Vaccinated Counties

13. Go to www.mydakotahis.com (Dashboard Home).
14. Click the "Measles" tab.
15. Scroll down to the "Lowest MMR Coverage Counties" chart.
16. Review the horizontal bar chart — counties are ranked from lowest to highest MMR coverage rate.
17. Counties displayed in red (below 70% coverage) are at Critical risk. Counties in orange (70–85%) are at High risk.
18. Note the county names and their coverage rates for reporting or outreach purposes.
19. For additional context, scroll further to the Disease Risk Score and Population Risk charts to see how risk compounds when low coverage meets larger populations.

5.3 How to Download Raw Data for a Specific Year and County

20. Go to www.mydakotahis.com/data (the Data Explorer page).
21. Click the dataset card you want (e.g., "Hepatitis A Cases").
22. Wait for the table to load — the "Total Records" stat in the hero banner will populate.
23. In the controls bar, select the desired year from the "Year" dropdown (e.g., "2023").
24. Select the desired county from the "County" dropdown (e.g., "Cass").
25. The results bar will show how many records match your filters (e.g., "Showing 4 of 1,084 records").
26. Click the "Download" button to open the Download Panel.
27. Click "Filtered CSV" to download only the records matching your selected year and county.

28. The file downloads immediately to your computer as a CSV file.

5.4 How to Compare Vaccination Coverage Across All Counties

29. Go to www.mydakotahis.com/data.
30. Click the "Vaccination Coverage" dataset card.
31. The table loads with 1,084 records covering all counties, years, and diseases.
32. In the controls bar, use the "Disease" filter to select a specific disease (e.g., "Measles").
33. Use the "Year" filter to select a specific year (e.g., "2025").
34. The table now shows one row per county with their 2025 Measles vaccination coverage percentage.
35. Click the "Coverage_Pct" column header to sort by coverage rate (ascending or descending).
36. Identify counties with the lowest coverage rates at the bottom when sorted ascending.
37. Download the filtered dataset using "Filtered CSV" for use in a report or further analysis.

5.5 How to Understand the Outbreak Timeline

38. Go to www.mydakotahis.com (Dashboard Home).
39. Click the "Measles" tab.
40. Review the Monthly Case Trend Chart — this shows case counts by month for 2025 and 2026. The spike in May 2025 (34 cases) and February 2026 (24 cases) represent the two largest waves.
41. Review the Annual Totals section: zero cases from 2022–2024, then 36 cases in 2025, and 24 cases through February 2026.
42. Cross-reference with the MMR Trend chart: as the vaccination rate dropped from 92.23% (2022) to 89.71% (2025), the outbreak emerged — demonstrating the direct relationship between declining coverage and disease resurgence.
43. Review the County Case Comparison chart to see which counties were most affected in each year.

6. Tips, Best Practices & Frequently Asked Questions

6.1 Tips for Getting the Most Out of the Dashboard

- Use the Dashboard Home for visual analysis (charts and maps) and the Data Explorer for data access (download and raw records).
- Always check which year filters are active before drawing conclusions. The "All Years" view may blend data from different outbreak periods.
- When reviewing maps, use the 2025/2026/All toggle buttons to understand how the outbreak's geographic distribution changed between years.
- Compare the MMR Rate Trend chart with the County Case chart to identify causal relationships between low vaccination and outbreak intensity.
- Use the active filter chips in the Data Explorer to keep track of what filters you have applied — you can remove individual filters by clicking the "x" on each chip.
- Sort the Vaccination Coverage table by Coverage_Pct (ascending) to quickly rank all counties from least to most vaccinated.

6.2 Frequently Asked Questions

Q: Is an account required to access the dashboard?

No. MyDakota HIS is fully publicly accessible. No login, registration, or payment is required for any feature, including data downloads.

Q: How current is the data?

Measles case data reflects confirmed case counts from ND HHS and is among the most current data on the site. Vaccination coverage data is updated annually. Other disease datasets (Hepatitis A, Rotavirus, RSV, Influenza) reflect ND HHS surveillance records and are updated periodically. Check the data source attribution on the Data Explorer hero banner for the stated coverage period (2022–2026).

Q: What is the 95% herd immunity threshold?

For Measles, public health experts worldwide recognize that at least 95% of the population must be immune (through vaccination or prior infection) to prevent community-level spread. When coverage drops below this threshold — as North Dakota's has, reaching 89.71% statewide by 2025 — the virus can find enough susceptible individuals to sustain an outbreak. Individual counties with rates as low as 57% (Sheridan County) are at severe risk.

Q: What does "Insufficient Data" mean on the risk charts?

"Insufficient Data" is displayed for counties where there is not enough historical case data or vaccination records to calculate a reliable risk score. This often applies to very small or very rural counties with small populations where year-to-year variation makes trend analysis unreliable.

Q: Can I share or embed the dashboard?

The dashboard is publicly accessible via its URL (www.mydakotahis.com), so you can freely share the link. For embedding in other websites or applications, contact the MyDakota HIS team through the Contact page.

Q: Is the Measles data real or simulated?

The Measles case data on MyDakota HIS reflects actual confirmed case counts from the North Dakota 2025–2026 Measles outbreak as reported by ND HHS. All other disease datasets (Hepatitis A, Rotavirus, RSV, Influenza) are generated from ND HHS surveillance data and may use modeled values for certain demographic breakdowns. The site clearly labels Measles data as "Real data" in the Data Explorer.

Q: The map is not loading — what should I do?

The interactive map requires a working internet connection to load map tiles from CartoDB and OpenStreetMap. If the map appears blank or fails to load: (1) Check your internet connection. (2) Disable any browser extensions that might block third-party content (such as ad blockers or script blockers). (3) Try refreshing the page. (4) Try a different browser (the dashboard is optimized for modern browsers like Chrome, Firefox, Edge, and Safari).

Q: Why does the Data Explorer show "—" for Total Records briefly?

The total record count is calculated dynamically when the page loads. For a brief moment (less than one second on most connections), it displays "—" as a placeholder while the data initializes. This is normal and expected behavior.

7. Technical Reference

7.1 Browser Compatibility

MyDakota HIS is designed for modern web browsers. The following browsers are fully supported:

Browser	Support
Google Chrome	Version 90 and above — recommended for the best experience.
Mozilla Firefox	Version 88 and above.
Microsoft Edge	Version 90 and above.
Apple Safari	Version 14 and above (macOS and iOS).
Samsung Internet	Version 14 and above (Android devices).

Note: *Internet Explorer is not supported. If you are using IE, please upgrade to a modern browser.*

7.2 Mobile and Tablet Compatibility

MyDakota HIS is fully responsive and works on smartphones and tablets. On smaller screens:

- The header navigation compresses and some subtitle text is hidden to save space.
- Dataset cards in the Data Explorer reflow into a 2-column grid instead of 5 columns.
- The controls bar stacks vertically for easier touch interaction.
- The data table enables touch-based horizontal scrolling.
- Pagination controls stack vertically.
- Charts resize proportionally to fit the screen width.

7.3 Technologies Used

The dashboard is built on modern open-source technologies:

Technology	Purpose
Leaflet.js (v1.9.4)	Powers the interactive county maps. Provides zoom, pan, tooltips, and GeoJSON rendering.
Chart.js (v4.4.0)	Powers all bar charts, line charts, and doughnut charts throughout the dashboard.
CartoDB / OpenStreetMap	Provides the map tile background used in the county maps.
Font Awesome (v6.4.0)	Provides the icons used throughout the interface (shield, virus, database, etc.).

DM Sans / DM Serif Display	Google Fonts typefaces used for the clean, professional visual design.
HTML5 + CSS3 + Vanilla JavaScript	All dashboard logic is built without a JavaScript framework, using plain web standards for maximum performance and compatibility.

7.4 Data Formats for Download

Format	Description
CSV	Comma-Separated Values. Opens in Excel, Google Sheets, R, Python (pandas), SPSS, and most statistical tools. All values are plain text with commas as delimiters. Strings containing commas are wrapped in quotes.
JSON	JavaScript Object Notation. A structured array of objects where each record is a JSON object with field names as keys. Preferred format for developers and API integrations.
Filtered CSV	Same format as CSV, but contains only rows matching the current filter settings. Useful for exporting a precise subset without post-processing.

7.5 GeoJSON County Data

The interactive maps use a built-in GeoJSON dataset representing all 53 North Dakota counties as simplified rectangular polygons. This GeoJSON is embedded directly in the dashboard page (no separate file download required), which means the maps load and render instantly without additional network requests.

8. Quick Reference Summary

8.1 Dashboard Home — Key Elements at a Glance

Element	Function
Tab Bar	Switch between Overview, Measles, Hepatitis A, Rotavirus, RSV, and Methodology views.
Filter Bar	Filter all dashboard data by Year and County.
4 Stat Cards	Show key metrics for the active disease tab (total cases, affected counties, critical/high risk counties).
Monthly Trend Chart	Line chart showing case counts by month, split by year.
County Case Chart	Horizontal bar chart ranking counties by case count.
MMR Trend Chart	Line chart showing statewide MMR vaccination rate over time.
Lowest MMR Chart	Bar chart of the 10 counties with lowest vaccination rates.
Risk Score Chart	Bar chart of the 10 counties with highest composite risk scores.
Risk Distribution Doughnut	Shows distribution of all 53 counties across risk levels.
Population Risk Chart	Risk scores weighted by population size.
Susceptible Population Chart	Estimated unvaccinated individuals by county (total and under 18).
County Map	Interactive map of North Dakota color-coded by case count with 2025/2026/All toggles.

8.2 Data Explorer — Key Elements at a Glance

Element	Function
Hero Banner	Shows total records, datasets count, counties count, and date range.
Dataset Cards	6 cards to select: Measles Cases, Hepatitis A Cases, Rotavirus Cases, RSV Cases, Influenza Cases, Vaccination Coverage.
Search Box	Real-time text search across county, gender, race, and disease fields.

Year Filter	Filter by year: 2022, 2023, 2024, 2025, or 2026.
County Filter	Filter to a specific North Dakota county.
Gender Filter	Filter to Male or Female (Case datasets only).
Race Filter	Filter to White, Black American, Hispanic, or Native American (Case datasets only).
Disease Filter	Filter by disease (Vaccination Coverage dataset only).
Clear Button	Reset all filters to defaults.
Download Button	Open download panel to export CSV, JSON, or Filtered CSV.
Results Bar	Shows "Showing X of Y records" and active filter chips.
Data Table	Sortable, paginated table with visual badges for gender, race, and coverage.
Pagination Controls	Navigate pages; choose rows per page (25, 50, 100, 250).
Measles Insight Charts	Special charts shown when Measles Cases dataset is active.

8.3 Disease Color Reference

Disease / Dataset	Color
Measles	Red (#c0392b)
Hepatitis A	Orange (#d35400)
Rotavirus	Green (#1e8449)
RSV	Blue (#2471a3)
Influenza	Purple (#7d3c98)
Vaccination Coverage	Dark Purple (#6c3483)

8.4 Risk Level Color Reference

Risk Level	Color & Score Range
Critical	Red — Score above 4.0
High	Orange — Score 2.5 to 4.0
Moderate	Yellow/Amber — Score 1.5 to 2.5
Minimal Risk	Green — Score below 1.5

Insufficient Data	Gray — Not enough data for scoring

— End of User Guide —

For questions, visit www.mydakotahis.com/contact

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